2.3 Applying Exponent Laws to Generate Equivalent Algebraic Expressions

Lesson Introduction

Benchmark	MA.912.NSO.1.2 Generate equivalent algebraic expressions using the properties of exponents.	Learning Target	• I can generate equivalent algebraic expressions using properties (laws) of exponents.

	Building On	Working Toward
Benchmark Coherence	MA.8.NSO.1.3 Extend previous understanding of the laws of exponents to include integer exponents. Apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to integer exponents and rational number bases, with procedural fluency. MA.8.AR.1.1 Apply the Laws of Exponents to generate equivalent algebraic expressions, limited to integer exponents and monomial bases.	MA.912.NSO.1.3 Generate equivalent algebraic expressions involving radicals or rational exponents using the properties of exponents.

Essential Questions	• What are the laws of exponents and how are they applied when writing equivalent expressions? • Are there always domain restrictions? If not, when would you have domain restrictions?
Lesson Narrative	In this lesson, students will determine equivalent expressions using the product and quotient exponent laws. During this lesson, students will need to apply their understanding of the previous two lessons in order to generate more complex equivalent expressions and prove that specific monomial expressions are equivalent using the laws of exponents.

Component Summary				
	2.3.1 Warm-Up (~5 min.)	Creating a true mathematical statement *(SE p. 68)*	2.3.4 Bring It Together (10 -15 min.)	Proving expressions are equivalent using laws of exponents *(SE p. 70)*
	2.3.2 Guided Practice (10 -15 min.)	Examining different methods for generating equivalent expressions using the laws of exponents *(SE p. 68)*	2.3.5 Your Turn! (10 min.)	Generating and proving expressions are equivalent using laws of exponents *(SE p. 71)*
	2.3.3 Try It (10 -15 min.)	Generating equivalent expressions using the laws of exponents *(SE p. 70)*	2.3.6 Wrap-Up (~5 min.)	Error Analysis of generating equivalent expressions using laws of exponents *(SE p. 72)*

	Glossary	Materials	Additional Preparation
Resources	Key Terms: N/A Additional Terms: N/A	• (Optional) Calculator	N/A

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may mix up the laws of exponents.
Students may raise a coefficient to the power that a variable is being raised to.
Students may struggle when working with negative exponents and applying the appropriate law.</td><td>Lessons 1 and 2 are largely a review from middle school content. This lesson is a culmination of that content that brings together the laws of exponents as it applies to Algebra 1.
Understanding domain restriction is not the learning goal of the lesson but it is important for future topics in mathematics.
BEST uses “properties” and “laws” interchangeably. Both laws of exponents and properties of exponents need to be used during instruction to ensure students understand they are the same.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Compare and Connect
2.3.2 Guided Practice positions students to compare different approaches and methods to generate equivalent expressions. Consider applying this routine to encourage mathematical conversation and foster meta-cognitive and meta-linguistic awareness.</td><td>MA.K12.MTR.2.1: Multiple Representations
Students will be working to generate equivalent expressions using laws of exponents. Utilize the progression of the lesson to highlight the process by which students are generating expressions and proving that two expressions are equivalent, making connections to expanded form for a conceptual understanding.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">Throughout the lesson, students will be individually analyzing and generating equivalent expressions. Utilize the built-in partner discussion in the different components of the lesson to position students to think deeply and to find different entry points into the content. Some students may need to make sense of the content and information individually first, while others might prefer to process the information verbally with a partner before formalizing their observations.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
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2.3.1 Warm-Up: Dividing Monomials

Component Overview: *Students will use the digits $1-10$ to fill in boxes to create a true mathematical statement.*

2.3.1 Warm-Up: Dividing Monomials

1. Using the digits 1 to 10, at most one time each, fill in the boxes to make the statement true.

$$\frac{\boxed{3}\cdot x^{\boxed{1}}\cdot y^{\boxed{7}}}{\boxed{4}\cdot y^{\boxed{5}}}=\frac{\boxed{6}\cdot x^{\boxed{10}}\cdot y^{\boxed{2}}}{\boxed{8}\cdot x^{\boxed{9}}}$$

After students have had a few minutes to determine an answer, consider having students write their possible mathematical statements in a list on the board.

Spend time asking students about the strategy they used to determine their coefficients and exponents.

2.3.2 Guided Practice: Using Different Methods to Generate Equivalent Expressions

Component Overview: *Students work to generate equivalent monomial expressions using different methods. Monitor students as they work, and use questioning if students need help finding an entry-point to a problem.*

2.3.2 Guided Practice: Using Different Methods to Generate Equivalent Expressions

1. Consider the expression $\frac{(4x)(7x^3)}{2x^2}$.

   a. Use expanded form to generate an equivalent expression for $\frac{(4x)(7x^3)}{2x^2}$, where $x \neq 0$.

   $$\frac{(4x)(7x^3)}{2x^2} = \frac{(4 \cdot x)(7 \cdot x \cdot x \quad x)}{(2)(x \cdot x)} = 14x^2$$

   b. For the expression shown, indicate the laws of exponents used to generate the equivalent expressions.

$\frac{(4x)(7x^3)}{2x^2} = \frac{28x^4}{2x^2}$ $= 14x^2$	☐ Negative Exponent ☐ Product of Powers ☐ Quotient of Powers ☐ Power of a Product ☐ Zero Exponent ☐ Power of a Power ☐ Power of a Quotient

Students are positioned to compare different approaches to generate equivalent expressions. Utilize the structure of the lesson to encourage students to compare the methods by each student and connect that approach to their own understanding of the laws of exponents.

2. Three students used different methods to generate an equivalent expression for the expression $\left(\frac{5a^3 \cdot 3a^4}{6a^2}\right)^2$, where $a \neq 0$.

   a. Finish the work for each student.

Sebastian	Ana
$\left(\frac{5a^3 \cdot 3a^4}{6a^2}\right)^2 =$	$\left(\frac{5a^3 \cdot 3a^4}{6a^2}\right)^2 =$
$\frac{25a^6 \cdot 9a^8}{36a^4} =$	$\left(\frac{15a^7}{6a^2}\right)^2 =$
$\frac{225a^{14}}{36a^4} = \frac{25a^{10}}{4}$	$\frac{225a^{14}}{36a^4} = \frac{25a^{10}}{4}$

Consider partnering students based on their strengths to provide a variety of entry points. Students whose strengths lie in making use of pattern or structure visually pair well with students whose strengths lie in describing meaningful context.

2.3.2 Guided Practice: Using Different Methods to Generate Equivalent Expressions (continued)

Libby
$\left(\frac{5a^3 \cdot 3a^4}{6a^2}\right)^2 =$
$\left(\frac{5a^3 \cdot 3a^4}{6a^2}\right)\left(\frac{5a^3 \cdot 3a^4}{6a^2}\right) =$
$\left(\frac{5 \cdot 5 \cdot \cancel{3} \cdot \cancel{3} \cdot a \cdot a \cdot a \cdot a \cdot a \cdot a \cdot a \cdot a \cdot a \cdot a \cdot \cancel{a \cdot a \cdot a \cdot a}}{\underset{2}{\cancel{6}} \cdot \underset{2}{\cancel{6}} \cdot \cancel{a \cdot a \cdot a \cdot a}}\right) =$
$\frac{25a^{10}}{4}$

b. Discuss with your partner the differences in the methods used by each student. Then, determine if the different methods each produced the same equivalent expression.

Answers will vary. Sebastian applied the Power of a Quotient law first and then Product of Powers law then simplified using the Quotient of Powers law. Ana simplified inside the parenthesis first using the Product of Powers law and then applied the Power of a Quotient law followed by the Quotient of Powers law. Libby expanded the expression first and then canceled from the numerator and denominator before simplifying.

After students discuss with their partners, consider asking the following:

- *How is the work started by Sebastian, Ana, and Libby similar? How is it different?*
- *What does that make you think about how to approach writing equivalent expressions?*
- *Is there an approach that you prefer? Why?*

2.3.3 Try It: Generating Equivalent Algebraic Expressions

Component Overview: *Students work independently to generate equivalent monomial expressions and then compare their work with their partner. Use questioning if needed to position students to determine the method needed to solve.*

2.3.3 Try It: Generating Equivalent Algebraic Expressions

1. For each of the following, find an equivalent expression.

$\frac{(-10h)^2}{5h^6}$ where $h \neq 0$	$\left(\frac{-2c \cdot -3c^4}{8c^2 \cdot -6c}\right)^2$ where $c \neq 0$	$\left(\frac{12ry^2 \cdot 3r^4y^5}{6r^7y^{10}}\right)$ where $r \neq 0$ and $y \neq 0$
$\frac{20}{h^4}$	$\frac{c^4}{64}$	$\frac{6}{r^2y^3}$

2. Compare your answers with your partner and resolve any differences, if necessary.
   Answers will vary.

3. Compare and discuss the methods you used to create each equivalent algebraic expression.
   Answers will vary.

Discussion provides auditory entry-points. Consider utilizing sentence stems and provide ample time for students to grapple with academic language.

- *I think the equivalent expression is _____ because …*
- *I used the ____ approach because …*
- *The exponent on _____ is _____ because …*
- *The coefficient is _____ because …*

Consider challenging students to also explain the domain restrictions for the problems presented in question 1.

2.3.4 Bring It Together: Generating Multiple Equivalent Algebraic Expressions

Component Overview: *Students use their understanding of the laws of exponents to generate and prove expressions are equivalent.*

2.3.4 Bring It Together: Generating Multiple Equivalent Algebraic Expressions

1. Discuss with your partner how to show $3^{(2x+4)}$ is equivalent to $81(9)^x$. Summarize your discussion.
Answers will vary. I can substitute a value in for x to show both expressions have the same value.

2. Prove $3^{(2x+4)} = 81(9)^x$ by filling in the value for each blank.

$$3^{(2x+4)} = 3^{(2x)} \cdot 3^{\boxed{4}}$$

$$3^{(2x)} \cdot 3^{\boxed{4}} = 3^{(2x)} \cdot 81$$

$$81 \cdot 3^{(2x)} = (3^2)^{\boxed{x}} \cdot 81$$

$$(3^2)^{\boxed{x}} \cdot 81 = (9)^{\boxed{x}} \cdot 81$$

$$3^{(2x+4)} = 81(9)^x$$

3. Prove $z^{(5y-6)} = \frac{(z^5)^y}{z^6}$ by using the laws of exponents. Show your work.
Answers will vary.
$z^{5y} \cdot z^{-6}$
$z^{5y} \cdot \frac{1}{z^6}$
$(z^5)^y \cdot \frac{1}{z^6}$
$\frac{(z^5)^y}{z^6}$

4. Use the laws of exponents to write two equivalent expressions for $1.2^{(4+2x)}$. Show your work.
Answers will vary.
$1.2^4 \cdot 1.2^{2x}$
$1.2^4 \cdot (1.2^2)^x$

For question 2, consider writing a justification for each step. Include the laws of exponents in the justification. This move will provide a visual connection when students do not know where to start.

Consider using different methods of presentation to best meet the students individually. Consider providing 3 options based on observations from Lessons 1 and 2.

- *Partner work*
- *Individual assignment*
- *Whole-group instruction*

2.3.5 Your Turn!

Component Overview: *Students use their understanding of the laws of exponents to generate and prove expressions are equivalent independently.*

2.3.5 Your Turn!

1. Write an equivalent expression for $3^{9r^2} \cdot 3^{9r}$ using the laws of exponents.
   3^{9r^2+9r}

2. Prove $2.1^{(3x+1)} = 2.1(9.261)^x$ by using the laws of exponents. Show your work.
   $2.1^{3x} \cdot 2.1^1$
   $2.1^{3x} \cdot 2.1$
   $(2.1^3)^x \cdot 2.1$
   $(9.261)^x \cdot 2.1$
   $2.1(9.261)^x$

3. Use the law of exponents to write two equivalent expressions for $6.2^{(2-3x)}$. Show your work.
   $\frac{6.2^2}{6.2^{3x}}$

   $\frac{38.44}{(6.2^3)^x} = \frac{38.44}{238.328^x}$

Probe students to make connections between equivalent expressions by utilizing the structure of the lesson. Consider asking questions such as:

- *In question 1, how does your expression compare with the given expression? What exponent law did you use?*
- *In question 2, what exponent law helps to prove that the two given expressions are equivalent?*

Consider challenging students to create their own question, like question 2, for another student to solve. Students should use at least two exponent laws when generating their question.

2.3.6 Wrap-Up: Error Analysis

Component Overview: *Students use their understanding of the laws of exponents to analyze and generate equivalent expressions independently. Consider using all or parts of this exercise as a formative assessment to gather data on student progress from this lesson.*

2.3.6 Wrap-Up: Error Analysis

1. Errors were made in finding expressions equivalent for $4^{(3s-1)}$. Identify the errors, then write two correct equivalent expressions.

$$4^{(3s-1)} = \frac{4^{(3s)}}{4^1} = \frac{(4^3)^s}{4^1} = \frac{64 \cdot 4^s}{4} = 4 \cdot 4^s$$

Answers will vary. The error was made in the next to last step. The 4^3 should be rewritten as 64 and then raised to the power of s. The simplest equivalent form is $\frac{64^s}{4}$.